Experiment 6 – Mode selection in laser diodes: DFB laser

Explanation: goal, principle, procedure and why we need it later

1. The experiment in one sentence

A normal diode laser (Fabry-Perot) oscillates on several modes; a DFB laser selects exactly one with a grating – we compare both with a Fabry-Perot spectrometer and measure how the wavelength changes with temperature and current.

2. Goal of the experiment

- Calibrate the Fabry-Perot spectrometer (sawtooth → time axis → frequency via the FSR of 10 GHz).
- Determine the line width of the DFB laser and compare it with the resolution (67 MHz).
- Determine $\Delta\lambda/\Delta T$ and $\Delta\lambda/\Delta I$ of the DFB laser from ≥ 10 values each.
- Analyse the spectrum of the Fabry-Perot laser: number of modes, mode spacing, mode hops.

3. Principle – the logic behind it

3.1 The Fabry-Perot resonator

Two mirrors a distance L apart (index n) transmit only waves that are in phase after one round trip ($\varphi = 4\pi nL/\lambda = 2\pi m$). The transmission follows the **Airy formula** $I_t/I_0 = (1-R)^2/(1+R^2-2R\cos\varphi)$. Maxima at $\nu_m = m\cdot c/(2nL)$, spacing **FSR = c/(2nL)**, sharpness **finesse $F = \pi\sqrt{R}/(1-R)$**, FWHM = FSR/F. The better the mirrors, the sharper the maxima.

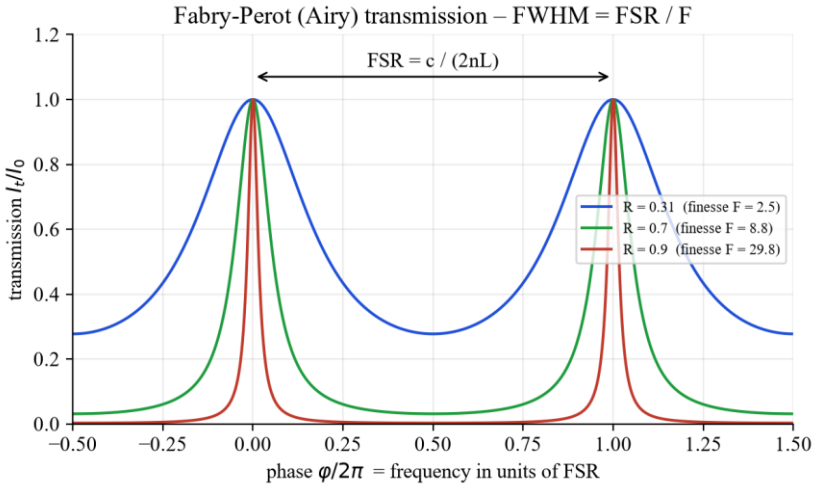


Fig. 1: Airy transmission for R = 0.31 / 0.7 / 0.9.

3.2 Fabry-Perot laser: many modes

The crystal facets (R ≈ 31 %) are not wavelength selective: all longitudinal modes with net gain oscillate, spacing $\Delta\nu = c/(2nL)$, $\Delta\lambda = \lambda^2/(2nL)$ (L = 1 mm, n = 3.5: 43 GHz = 0.34 nm). Transverse modes are suppressed by the waveguide (TE_{00}).

3.3 Bragg grating, DBR and DFB

A grating of period Λ reflects constructively only at $\lambda_B = 2\, n_{eff}\, \Lambda / m$ ("the wavelength has to fit into the grating"). **DBR:** a grating outside the gain region replaces one mirror. **DFB:** the refractive index (index coupling) or the gain (gain coupling) is periodically modulated in the active zone itself. The allowed mode spacing becomes larger by the factor L/L_{Bragg} – ideally only one mode lies in the gain range, side modes are suppressed.

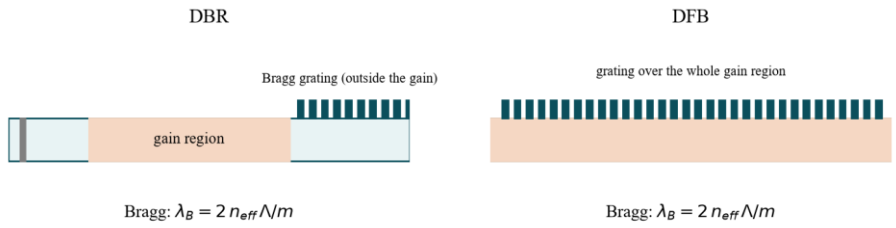


Fig. 2: DBR and DFB structure (schematic).

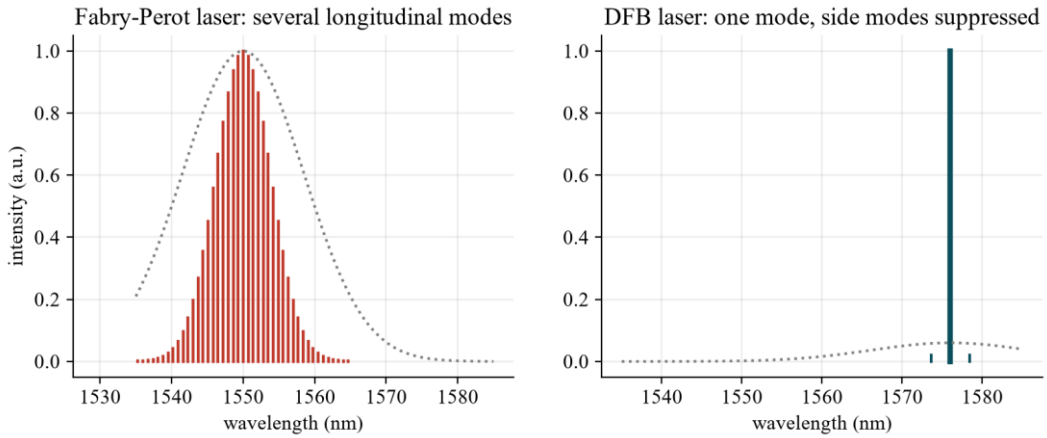


Fig. 3: Spectrum of the FP laser (many modes) and the DFB laser (one mode).

3.4 Wavelength tuning

FP laser: the gain maximum moves quickly with T (≈ 0.4–0.5 nm/K at 1550 nm), the modes slowly → **mode hops**. **DFB laser:** only n_{eff}(T, carrier density) changes: $d\lambda/dT = \lambda_B \cdot (dn_{eff}/dT)/n_{eff} \approx 0.1$ nm/K (≈ 12 GHz/K), smooth and without jumps. The current acts through heating ($\propto I^2$) and carrier density; the grating fixes the coarse wavelength.

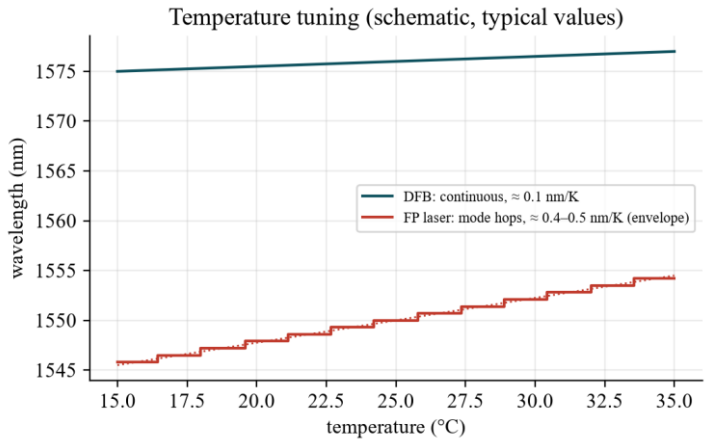


Fig. 4: Tuning with temperature (schematic, typical values).

3.5 The Fabry-Perot spectrometer

A Fabry-Perot resonator with **piezo-adjustable length** followed by an InGaAs photodiode: a sawtooth voltage (≈ 10 V, 100 Hz) sweeps L – the time axis becomes the wavelength axis. When L changes by λ/2 the same laser line is transmitted again (m → m+1): **the distance between two lines = FSR = 10 GHz** (calibration). Resolution 67 MHz, finesse ≈ 150. Conversion: **$\Delta\lambda/\lambda = \Delta\nu/\nu$** (from $\nu = c/\lambda$), never $\Delta\nu = c/\Delta\lambda$. At 1550 nm: 10 GHz = 80 pm, 67 MHz = 0.54 pm.

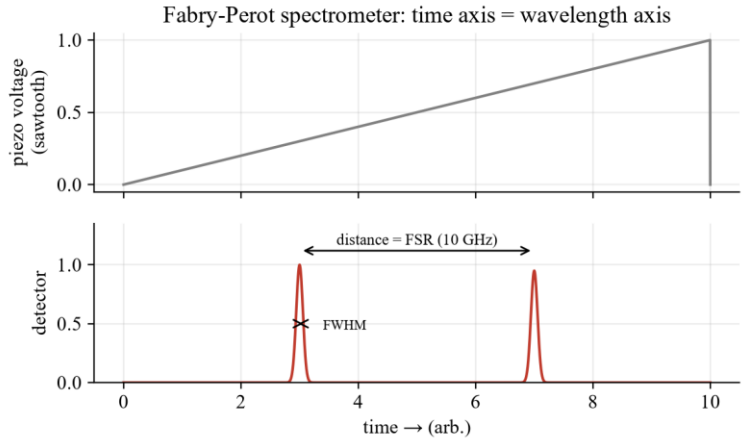


Fig. 5: Sawtooth and detector signal – two lines = one FSR.

4. Procedure step by step

Step	What to do?	Evaluation
1	DFB laser on the current/temperature controller (constant current: knob left, ILD, on, raise slowly; lasing from ≈ 20 mA), couple into the FP spectrometer; adjust for the narrowest lines	—
2	Choose the sawtooth amplitude so that two lines are visible	Calibration: 10 GHz/Δt_{lines}
3	Read the FWHM of the DFB line	$\Delta\nu$ = calibration factor · $\Delta t_{½}$; compare with 67 MHz
4	Vary T (constant current), ≥ 10 values; then vary I (constant T), ≥ 10 values; do not exceed the FSR	$\Delta\lambda = \lambda^2\Delta\nu/c$; slope $\Delta\lambda/\Delta T$, $\Delta\lambda/\Delta I$
5	Couple in the FP laser: describe the spectrum, look for mode hops, count modes	Mode spacing; for L = 1 mm: 42.8 GHz = 0.34 nm (> FSR → orders overlap)

Optional: LTspice demo of the Airy function: FabryPerot_Airy_Transmission.asc (or *04_LTspice/LTspice_starten.bat*).

5. What to watch in the lab

- **Invisible IR radiation (1550/1576 nm), up to 2.5 mW: laser goggles, never look into the beam.**
- Set the temperature via the sensor resistance (kΩ, calibration curve) and wait until it is stable.
- The DFB line moves at ≈ 12 GHz/K, faster than the FSR (10 GHz): use small temperature steps or keep count of the orders.
- Use only $\Delta\lambda/\lambda = \Delta\nu/\nu$ – never $\Delta\nu = c/\Delta\lambda$.
- Protocol: expectation – measurement – error analysis for every point.

6. Why do we need this later?

- **Fibre communication:** DFB lasers are the standard sources (narrow line, stable wavelength, wavelength-division multiplexing).
- **Sensing and spectroscopy:** gas measurement (absorption lines) needs tunable single-mode lasers.
- **Experiment 5 → 6:** FP laser (threshold, mode hops) ↔ DFB laser (one mode, continuous tuning).
- **Metrology:** the Fabry-Perot interferometer is a tool with extremely high spectral resolution.

7. Typical questions – short answers

Question	Short answer
Airy formula, FSR, finesse?	$I_t/I_0 = (1-R)^2/(1+R^2-2R\cos\varphi)$; FSR = c/(2nL); $F = \pi\sqrt{R}/(1-R)$; FWHM = FSR/F.
Why many modes in the FP laser?	Facets are not wavelength selective; all modes with net gain oscillate.

How does the DFB laser get one mode?	Grating in the gain region: only $\lambda_B = 2n_{eff}\Lambda/m$ is fed back.
DBR vs. DFB?	DBR: grating outside the gain; DFB: in the active region.
Index vs. gain coupling?	Real part n modulated (reflection) vs. gain g modulated.
Tuning DFB vs. FP?	DFB ≈ 0.1 nm/K smooth; FP ≈ 0.4–0.5 nm/K with jumps.
Why two lines = FSR?	Change of L by λ/2 → next order m+1 → the same line again.
Resolution/finesse of the device?	67 MHz; FSR 10 GHz; F = 149.
$\Delta\lambda \leftrightarrow \Delta\nu$?	$\Delta\lambda/\lambda = \Delta\nu/\nu$; 10 GHz ≈ 80 pm at 1550 nm.
Mode spacing for L = 1 mm?	c/(2nL) = 42.8 GHz = 0.34 nm; > FSR → aliasing in the FP spectrometer.
Line width of the DFB?	Typically 1–10 MHz < resolution: the measured value is an upper bound.
Safety?	Invisible IR: laser goggles, never look into the beam.

Date: **16 Dec 2026**, pre-test (Antestat) 14:00, building BA, room BA 114 (group 9). Note: the English manual is marked “preliminary”.